

Problem

It is expected that a parallel RLC resonant circuit has a midband admittance of $25 \times 10^{-3} \text{ S}$, quality factor of 120, and a resonant frequency of 200 krad/s. Calculate the values of R , L , and C . Find the bandwidth and the half-power frequencies.

Step-by-step solution

Step 1 of 4

Calculate the bandwidth of parallel circuit.

$$B = \frac{\omega_0}{Q}$$

Substitute 200×10^3 for ω_0 and 120 for Q .

$$\begin{aligned} B &= \frac{\omega_0}{Q} \\ &= \frac{200 \times 10^3}{120} \\ &= 1.667 \text{ krad/s} \end{aligned}$$

Step 2 of 4

Calculate the half power frequencies.

$$\begin{aligned} \omega_1 &= \omega_0 - \frac{B}{2} \\ &= 200 \times 10^3 - \frac{1667}{2} \\ &= 199.166 \text{ krad/s} \end{aligned}$$

And,

$$\begin{aligned} \omega_2 &= \omega_0 + \frac{B}{2} \\ &= 200 \times 10^3 + \frac{1667}{2} \\ &= 200.833 \text{ krad/s} \end{aligned}$$

Therefore, the half-power frequencies are 199.166 krad/s and 200.833 krad/s.

Step 3 of 4

Write the formula for resonant frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$200 \times 10^3 = \frac{1}{\sqrt{LC}}$$

$$LC = \left(\frac{1}{2 \times 10^5} \right)^2$$

$$LC = 0.25 \times 10^{-10} \dots\dots (1)$$

Write the formula for quality factor.

$$Q = \frac{R}{\omega_0 L}$$

$$R = Q \omega_0 L$$

$$= (120)(200 \times 10^3) L$$

$$R = (24 \times 10^6) L \dots\dots (2)$$

Step 4 of 4

In a parallel RLC circuit the impedance is maximum and admittance is minimum. To get minimum admittance value choose, $R = 10 \text{ k}\Omega$.

From equation (2),

$$10 \times 10^3 = (24 \times 10^6) L$$

$$L = 0.42 \text{ mH}$$

Substitute 0.42 mH for L in equation (1).

$$(0.42 \times 10^{-3}) C = 0.25 \times 10^{-10}$$

$$C = 59.52 \text{ nF}$$

Hence, the parameters of RLC parallel network are $10 \text{ k}\Omega$, 0.42 mH and 59.52 nF .